Software Test Plan (STP) - Student Attendance Management System

Project: Student Attendance Management System
Version: 1.0
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1. Introduction

Purpose: This document defines the test plan for the Student Attendance Management System (SAMS) v1.0. It outlines the testing objectives, scope, strategy, resources, schedule, responsibilities, risks, and traceability approach.

Scope: Testing covers user authentication, role-based access, student and faculty management, subject/class management, attendance marking and modification, attendance viewing, percentage calculation, low-attendance identification, reports, notifications, and audit-related functions.

References: SAMS SRS v1.0, system design specifications, database design, UML use-case diagrams, and project requirements.

Definitions: SAMS (Student Attendance Management System), SRS (Software Requirements Specification), RTM (Requirements Traceability Matrix), FR (Functional Requirement), NFR (Non-Functional Requirement), UAT (User Acceptance Testing).

2. Test Items

- Authentication and authorization module
- Student management module
- Faculty and subject management module
- Class and enrollment management module
- Attendance management module
- Attendance calculation and monitoring module
- Reports and notifications module
- Admin, Faculty, and Student dashboards

3. Features to be Tested

Features mapped to SRS requirement IDs:

- SAM-F-001 to SAM-F-004: User authentication, invalid login handling, role-based access, and logout
- SAM-F-005 to SAM-F-011: Student, faculty, subject, class, assignment, and enrollment management
- SAM-F-012 to SAM-F-016: Attendance marking, authorization, modification, duplicate prevention, and record storage
- SAM-F-017 to SAM-F-020: Attendance viewing, percentage calculation, faculty access, and low-attendance identification
- SAM-F-021 to SAM-F-024: Reports, summaries, notifications, and attendance history
- SAM-NF-001: Response time within 3 seconds for at least 95% of normal requests
- SAM-NF-002: 99% monthly availability excluding scheduled maintenance
- SAM-NF-003: Role-based security
- SAM-NF-004: Usability
- SAM-NF-005: Data integrity
- SAM-NF-006: Auditability
- SAM-NF-007: Support for at least 100 concurrent users
- SAM-SR-001 to SAM-SR-005: Authentication, authorization, password hashing, HTTPS, and security logging

4. Features Not to be Tested

- Examination, fee, hostel, and learning-management functions outside the attendance system scope
- Third-party services that are not integrated into the project build
- Production infrastructure administration outside the application's responsibility
- Hardware-specific attendance devices, such as future QR or biometric hardware

5. Test Approach / Strategy

Levels:

- Unit testing (module-level functions and components)
- Integration testing (application, database, and module interactions)
- System testing (end-to-end SAMS functionality)
- Acceptance testing (faculty, student, and administrator workflows)

Types:

- Functional testing
- Regression testing
- Performance and load testing
- Usability testing
- Security testing
- Data integrity and validation testing

Entry Criteria: A stable build is available, test data is prepared, the test environment is operational, and the required modules are accessible.

Exit Criteria: 100% of planned test cases are executed, no critical defects remain open, and all defined acceptance criteria are satisfied or formally accepted.

5.1 Security Validation

- Validate login and role-based authorization for Admin, Faculty, and Student users
- Verify that passwords are stored using secure one-way hashing and not plaintext
- Verify HTTPS/TLS is enabled in the deployed environment
- Test unauthorized access to attendance and administrative functions

- Verify important attendance and administrative actions are logged with user and timestamp where applicable
- Perform input validation and security testing on login, attendance, and management forms

6. Test Environment

- Hardware: Desktop or laptop computers; mobile/tablet browsers may be used if responsive access is supported.
- Software: Windows/Linux/macOS, Python, Django, supported web browser, and MySQL database.
- Tools: Browser developer tools, Django testing framework/Python unit-testing tools, Postman where APIs are exposed, JMeter or similar load-testing tool, and Jira or another defect-tracking tool.
- Test Data: Dummy Admin, Faculty, and Student accounts; sample subjects/classes; enrolled students; attendance records containing Present and Absent statuses; and low-attendance scenarios.

7. Test Schedule

Milestones:

- Test case design: 16-Sep-2026
- Test environment setup: 17-Sep-2026
- Test execution start: 18-Sep-2026
- Functional and integration testing: 18-Sep-2026 to 24-Sep-2026
- Performance and security testing: 25-Sep-2026 to 27-Sep-2026
- UAT and final verification: 28-Sep-2026 to 30-Sep-2026

8. Test Deliverables

- Software Test Plan
- Test Cases
- Test Scripts
- Test Data

- Test Execution Logs
- Defect Reports
- Requirements Traceability Matrix
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Shivakumar	Prepare test plan, coordinate execution, review results
Test Engineer	Amit	Design and execute test cases, record results, log defects
Developer	Riona Jeena Dsouza	Support defect analysis, fixes, and regression testing
Product Owner / Team Lead	Atul Kandiyil	Review test results and approve readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early builds and perform smoke testing before full execution
Test environment downtime	Maintain a backup local or cloud test environment
Incomplete or inconsistent test data	Prepare and validate dummy test data before execution
Database errors or data loss	Maintain test database backups and use controlled test data
Changes to requirements during testing	Review requirement changes and update test cases and RTM

11. Assumptions & Dependencies

- The Django application and required dependencies will be available before test execution
- The test database will be accessible and contain valid sample data
- Admin, Faculty, and Student test accounts will be provided
- Required browsers and development/test tools will be available

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment is unavailable for more than 4 hours
- The build is too unstable and blocks more than 30% of planned test cases
- A critical database or authentication failure prevents meaningful testing

Resume testing if:

- Blocking or critical defects are resolved or an approved workaround is available
- The test environment is stabilized
- A new stable build is deployed and smoke testing passes

13. Test Case Management & Traceability

The Requirements Traceability Matrix (RTM) ensures that each SRS requirement is mapped to one or more test cases. Test cases shall use unique IDs and their execution results shall be recorded.

Examples:

SAM-F-001 (User login) → TC-AUTH-01

SAM-F-012 (Mark attendance) → TC-ATT-01

SAM-F-015 (Prevent duplicate attendance) → TC-ATT-04

SAM-F-018 (Attendance percentage) → TC-CALC-01

SAM-NF-001 (Response time) → TC-PERF-01

SAM-SR-003 (Password hashing) → TC-SEC-03

14. Test Metrics & Reporting

Metrics collected:

- Percentage of test cases executed
- Percentage of test cases passed and failed
- Number and severity of defects
- Defect aging
- Requirement coverage
- Regression test pass rate

Reports:

- Daily or periodic test execution status
- Defect status report
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Shivakumar	16-09-2026
Dev Lead	Riona Jeena Dsouza	16-09-2026
Product Owner	Atul Kandiyil	16-09-2026